Filling matrix spaces by transformed sets of arbitrary positive density exponent

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Abstract

We prove the matrix-set filling conjecture of Gill, Lifshitz, Pyber and Szabó for every positive size exponent, every finite field and every matrix dimension. The many-rank point-mass theorem proved in the companion note, together with the source's additive-energy and growth lemmas, gives an explicit number of summands depending only on the size exponent. All small ambient spaces are included.

Task 1: the matrix-set filling theorem

Write $A = \mathrm{Mat}_{n\times n}(\mathbb{F}_q)$ and $Q = |A| = q^{n^2}$, where q is a prime power and $n \geq 1$. For subsets, sums denote Minkowski sums. The following proves Conjecture 5.4 of [1].

Theorem 1. *For every $c > 0$ there is an integer $\mu(c)$ such that for any $X_1, \dots, X_{\mu(c)} \subseteq A$ with $|X_i| \geq Q^c$, there exist $a_i, b_i \in \mathrm{GL}(n, q)$ satisfying*

$$a_1 X_1 b_1 + \cdots + a_{\mu(c)} X_{\mu(c)} b_{\mu(c)} = A.$$

For $0 < c \leq 1$ one may take

$$D = 2^{\max\{15, \lceil 8/c \rceil\}}, \qquad \ell = \lceil 16/c \rceil, \qquad h = \ell + \lfloor 4/c \rfloor,$$
$$\nu = h 2^{h+1}, \qquad \mu(c) = 2D\nu.$$

For $c > 1$ the hypotheses are impossible, so $\mu(c) = 1$ suffices.

The sets may be different and need not contain zero or be symmetric. The source's established exponent $c = 9/10$ is its Theorem 5.2, equivalently Theorem 5.28; that case and the two lemmas used below are prior results, not new claims of this note.

The inputs and their exact scope

Let $T = \mathrm{GL}(n, q) \times \mathrm{GL}(n, q)$ act on the additive group A by $(g, h)x = gxh^{-1}$. This is an action by group automorphisms. We use the following two results from the completely read primary source.

Lemma 1 ([1], Corollary 5.13)**.** *Let B be a finite abelian group and $T \leq \mathrm{Aut}(B)$. For any nonempty subsets $Y_1, \dots, Y_L$ there exist $s_i \in T$ such that either*

$$\left| \sum_{i=1}^{L} s_i Y_i \right| \geq \frac{L}{2} \min_i |Y_i|,$$

or that sum is a union of cosets of a T-invariant subgroup N with $|N| \geq \min_i |Y_i|$.

Lemma 2 ([1], Corollary 5.17)**.** *Suppose B is finite abelian, $T \le \mathrm{Aut}(B)$ is finite, $\alpha, \beta \ge 1$, and $V \subseteq B$ has $|V| \le \beta$. Suppose for every $a \in B$ and $b_1, \dots, b_\ell \notin V$,*

$$\#\{(t_1, \dots, t_\ell) \in T^\ell : a = t_1 b_1 + \cdots + t_\ell b_\ell\} \le |T|^\ell / \alpha.$$

For $\delta > 0$, set $h = \ell + \lfloor 1/\delta \rfloor$ and $\nu = h 2^{h+1}$. If $Y_1, \dots, Y_\nu$ have size at least $\alpha^\delta \beta$, then there exist $s_i \in T$ such that their transformed sum either has size at least α, or contains a coset of a T-invariant subgroup N with $|N| \ge \min_i |Y_i|$.

These inputs are proved in the source using Olson's growth theorem and additive energy. In particular, they apply to unequal sets and to cosets, and their parameters have no dependence on $|B|$.

The new input is the companion point-mass theorem [2]: independent uniform matrices of ranks $r_1, \dots, r_k$ have each point probability at most C/Q, where $C = 32768$, whenever $\sum_i r_i \ge 2n$. For clarity, its mechanism is rectangular factorization: the probability that an $n \times r$ uniform factor is full rank is at least $4^{-r/n}$. Choose a rank prefix with total $2n \le R < 3n$. Conditioning both factors in each block costs at most 16^3, while the unconditioned product of an $n \times R$ and an $R \times n$ uniform matrix has point mass at most $8/Q$. Further independent summands do not increase the maximum. The full proof, exact verification, and fixed publication version of this dependency are linked in [2] and the accompanying audit.

Invariant subgroups and the exceptional ranks

The only T-invariant additive subgroups of A are $\{0\}$ and A. Indeed, a nonzero invariant subgroup contains, by left-right equivalence, $J_r = \mathrm{diag}(I_r, 0)$ for some $r \ge 1$. When $n \ge 2$, invariance and subtraction give

$$(I_n + E_{21}) J_r - J_r = E_{21}.$$

Its left-right orbit consists of every rank-one matrix. These include all scalar multiples of matrix units and additively generate A. When $n = 1$, the orbit of any nonzero element already contains all nonzero field elements. This also proves the assertion over nonprime fields without assuming in advance that an additive subgroup is field-linear.

Set $\eta = c/8$ and let V consist of matrices of rank less than ηn. The elementary rank-ball estimate in Fact 5.23(d) of [1] gives

$$|V| \le 4q^{2\eta n^2} = 4Q^{c/4}.$$

For completeness, if s is the largest integer less than ηn, every rank-at-most-s matrix has image in some s-dimensional subspace. There are at most $4q^{s(n-s)}$ such subspaces and at most q^{ns} maps into each, giving at most $4q^{s(2n-s)} \le 4q^{2\eta n^2}$. Here $s \ge 0$ because $c > 0$, and $\eta < 1$ for the range under consideration.

For fixed $b_i \notin V$, independently uniform $t_i \in T$ send b_i to independent uniform matrices of ranks at least ηn. Since $\ell = \lceil 16/c \rceil$ gives $\ell\eta \ge 2$, the point-mass theorem implies

$$\#\{(t_i) \in T^\ell : \sum_i t_i b_i = a\} \le C|T|^\ell / Q. \tag{1}$$

Uniformity here is the elementary orbit-stabilizer fact; no independence of matrix entries within a fixed rank class is assumed.

The two size regimes

Proof of Theorem 1. It remains to consider $0 < c \le 1$. Every X_i has at least two elements, since $Q^c > 1$ and its cardinality is an integer.

First suppose $Q \ge D$. Apply Lemma 2 with

$$\alpha = Q/D \ge 1, \qquad \beta = 4Q^{c/4} \ge 1, \qquad \delta = c/4.$$

Its orbit-count hypothesis follows from (1), because $D \ge C$. The size condition holds since $D^{c/4} \ge 4$ and

$$\alpha^\delta \beta = 4D^{-c/4}Q^{c/2} \le Q^{c/2} \le Q^c \le |X_i|.$$

Partition the $\mu = 2D\nu$ sets into $2D$ blocks of ν sets each. Lemma 2 transforms each block into a sum Y_j of size at least α. In its subgroup alternative, the subgroup has at least two elements and therefore equals A, so that block already fills A. In either case $|Y_j| \ge \alpha$, and also $|Y_j| \ge 2$ because a nonempty sumset contains a translate of each summand.

Now apply Lemma 1 to $Y_1, \dots, Y_{2D}$. Its first alternative gives size at least $D \min_j |Y_j| \ge D\alpha = Q$. Its subgroup alternative also gives all of A, since the subgroup has size at least two. Thus transformed block sums fill A. Composing their outer transformations with the inner transformations gives one element of T for each original set. Writing these as $a_i X_i b_i$ yields the required form, since an inverse of an invertible right multiplier is again an allowed right multiplier.

If $Q < D$, apply Lemma 1 directly to the μ original sets. The size alternative gives at least $\mu \ge Q$ elements, while the subgroup alternative again forces the whole group because $\min_i |X_i| \ge 2$. This proves the result in every remaining dimension and field, including all small ambient spaces. □

This is a separate set-filling assertion. Its proof explicitly uses the point-mass theorem of [2]; that dependency is not claimed as a second new point-mass result.

Proposed follow-up Tasks 2–5

These are proposed tasks, not additional completed results.

2. Determine the optimal growth rate of the least admissible $\mu(c)$ as $c \downarrow 0$, and compare it with the explicit bound above.

3. Give an algorithm for constructing the left and right multipliers, with a complexity bound in terms of the input sets and field size.

4. Establish the corresponding filling theorem for rectangular matrix spaces, with constants uniform in both dimensions and their ratio.

5. Characterize the subsets for which left multipliers alone suffice, and identify all invariant-subspace obstructions to a uniform theorem.

References

[1] N. Gill, N. Lifshitz, L. Pyber and E. Szabó, *Initiating the proof of the Liebeck–Nikolov–Shalev conjecture*, arXiv:2408.07800v2, 2024. Conjecture 5.4, Corollaries 5.13 and 5.17, and Theorem 5.28.

[2] Principia Math In Progress, *A uniform point-mass bound for sums of prescribed-rank matrices*, 22 September 2026, MathDB 364356, Theorem 1. Fixed published proof version.